

Types of data



1 State whether the following data are numerical or categorical:

- | | |
|---------------------------------|-------------------------------|
| a Amount owing on layby. | b Types of vegetables. |
| c Brands of tablets. | d Types of folders. |
| e Colours of folders. | f Brands of phones. |
| g Maximum snowfall. | h Daily UV index. |
| i Types of dogs. | j Favourite colours. |
| k Cost of a PS4 game. | l Types of fruits. |

2 State whether the following examples of numerical data are discrete or continuous:

- a** The number of classrooms in your school.
- b** Daily humidity.
- c** The ages of a group of people.
- d** The time taken to run 200 metres.
- e** Amount of money paid for walking the dog.
- f** The number of people taller than you in your class.
- g** The top speed of different models of motorcycles.
- h** The size of people's rare coin collections.
- i** Lengths of cats' whiskers.
- j** The distance from the Earth to nearby galaxies.
- k** How long it takes you to fall asleep at night.
- l** The amount of soft drink you drink in a day.
- m** The number of cans of soft drink you drink in a day.
- n** The world record time for solving ten Rubik's cubes.
- o** The hourly pay rate for a particular job.
- p** The rainfall in your city.
- q** Number of children in your sports team.
- r** Amount of money owing on a loan.
- s** Your marks in your most recent Science test.
- t** The number of views that a YouTube channel has.
- u** The strength of an earthquake

3 For each of the following survey questions, classify the answer as categorical, discrete numerical or continuous numerical data:



- a How far do you travel to school each day?
- b Where was your most recent holiday?
- c How many siblings do you have?
- d What modes of transport do you use to get to school?
- e How long did you sleep for last night?
- f How many people live in your house?
- g How long does it take for you to get to school everyday?
- h What languages do you speak at home?
- i What kinds of pets do you own?
- j How many pets do you own?
- k How many times have you broken your arm?
- l How long did you spend on homework last night?
- m How tall are you?
- n How much does your schoolbag weigh?
- o How many apps do you have in your phone?
- p Do you think recess is long enough?

Tables

4 Consider the given data set that shows the distance to the nearest hospital (in km):

26.4, 28.7, 15.1, 37.5, 11.8, 7.9, 16.7, 6.9, 5, 24.9, 35.3, 45.5

- a Is the data type continuous or discrete?
- b What is the farthest distance to the nearest hospital?

5 Glen surveys his class about their favourite movie genre. His results are shown in the table:

Genre	Drama	Comedy	Action	Horror
Students	5	7	6	2

- a What type of data is displayed in the table?
- b Which genre is the most popular among his class?
- c How many people did Glen survey?



- 6 Ten people were asked about the number of hours they spent outside last weekend. The results of the survey are shown:

5, 9, 9, 4, 6, 9, 4, 9, 6, 4

- a What type of data is this?
- b How many people spent exactly 6 hours outside?
- c What was the longest that anyone spent outside?
- d What was the most common amount of time spent outside?

- 7 Ten people were asked about the number and type of pets they own. The results of the survey are shown below:

Twelve fish	Three turtles	Two cats	Two cats	One dog
One dog	One dog, One cat	One cat	Five ferrets	One dog

- a What type of data is the type of pet?
- b How many cats do these ten people own altogether?
- c How many pets do these ten people own altogether?
- d What is the most common survey response?

Question biases

- 8 State whether the following questions are biased or fair?
- a Do you want a soup for lunch or a sandwich?
 - b What do you do on a Sunday morning?
 - c Do you like these shoes?
 - d Do you prefer this shirt or the one on the shelves at the moment?
 - e Do you think the government should be allowed to cut down some of the oldest trees in the area to construct a metro railway line in the city?
 - f Do you prefer newspapers or news on television?
 - g Do you prefer the full time degree program or part time degree program?
 - h Should the government enforce a minimum drinking age for its citizens?
 - i Do you eat at least the recommended number of servings of fruits and vegetables to ensure a healthy and long life?
 - j Do you think bike helmets should be mandatory for all bike riders?

l Do you exercise regularly?



m Do you feel that the TV news is an inaccurate portrayal of life's problems?

n Don't you think this newspaper is biased?

o Do you prefer the look and feel of thick lush carpeting in your living room?

p Do you take these extra strength multi-vitamins to supplement your diet?

9 Explain why the following survey questions are poor:

a The Prime Minister believes that taxes are too high, do you think taxes are too high?

b Do you have a Snapchat or Twitter account?

c Do your siblings ever drive you to school?

d Do you often waste time on social media?

e What brand of shoe polish do you use everyday?

10 Patricia surveys her class about their favourite music. Her results are shown in the table:

Genre	Country	Pop
No. Students	15	2

a According to the survey, which genre is most popular among her class?

b This was her survey question:

"The coolest kids like Country music, and nobody likes Pop. Do you like Country music or Pop music?"

Do you trust the results of Patricia's survey? Explain your answer.